WHAT HIDES MUSICAL GEOMETRY IN AUDIO REPRESENTATIONS?

Junyoung Koh

School of Music, Theatre and Dance, University of Michigan
Department of Artificial Intelligence, Yonsei University

ABSTRACT

Probing studies establish that audio representations encode musical attributes such as key, and this is routinely read as evidence that the representations are musically organised. We show these are different claims. Measuring the same 24 musical keys on the same corpus three ways, by linear decodability, by the geometry of class centroids, and by the geometry of a linear readout's weights, we find they dissociate. A log-CQT decodes key better than MERT-v1-330M, at .492 against .456, while its centroid geometry shows no circle-of-fifths structure at all, yet the same structure is strongly present in its probe weights. We then identify two empirically distinct regimes in which musical structure is hidden from the ambient metric. In the first, class-independent variation obscures centroid geometry at small sample sizes, and increasing the number of examples behind each centroid changes fifths geometry by a factor of 13.4, against only 1.7 for a thousandfold longer pooling window. In the second, averaging saturates without recovering octave equivalence at any sample size, whereas a rank-4 projection fitted only on GiantSteps keys, never on notes, exposes octave structure in individual NSynth recordings using under 0.3% of the variance. This distinction suggests a more general caution for interpreting representation geometry.

Index Terms— music representation learning, probing, representation geometry, key detection, octave equivalence

1. INTRODUCTION

Self-supervised audio encoders support strong linear probes [1] for key, pitch, genre and instrument [2, 3], and the usual inference is that the attribute is *organised* in the representation. Music makes that inference unusually testable, because music theory specifies not just which classes exist but how they are related. Keys a perfect fifth apart share six of seven scale degrees and sit adjacent on the circle of fifths, and pitches an octave apart are the same pitch class. A representation can therefore encode key perfectly while placing C and G no closer together than C and F♯.

We ask when that theoretical structure is visible in the geometry a downstream model actually consumes, and find that decodability, ambient geometry and readout geometry come apart (Sec. 4.1). We then separate two reasons a representation can hold musical structure that its metric does not show. In the first, the variation that distinguishes items of the same class is unrelated to the class, so it cancels once enough examples are pooled and the structure appears with no change to the representation. In the second, the variation tracks the class itself, so it is present in every example, survives any amount of averaging, and yields only to a change of metric. We show that centroid sample size is the dominant source of the apparent clip-level key effect (Sec. 4.2) and that note-level octave equivalence falls in the second regime (Sec. 4.3).

2. RELATED WORK

Linear probes are the standard instrument for asking what a representation encodes [1], and music audio encoders are routinely evaluated this way, with MERT [2] and MARBLE [3] reporting probe accuracy for key, pitch, genre and instrument. Hewitt and Liang [4] show that a probe's success has to be read against a control task that removes the structure of interest, and we adopt that design for representation geometry rather than for probe complexity.

The relational structure we test against is the one music cognition established for human listeners, in which key relations follow the circle of fifths and Krumhansl–Kessler profiles [5, 6] and pitch separates into height and chroma [7]. Chroma features [8, 9] build octave equivalence into the input by construction, which makes them a positive control rather than a competitor, while tonality-aware objectives such as STONE [10] impose key equivariance during training, whereas the question here is what a representation trained without such a constraint already contains. We also include the codec latent that music generation systems consume [11, 12].

3. MEASUREMENTS

Two regimes of hiding. Write an item of class y as

$$z = s_y + n_y + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid y] = 0, \tag{1}$$

Code, pre-registrations and all run artefacts: `https://github.com/Junst/topo-music`. Every threshold and kill criterion reported here was registered before the corresponding run.

with s_y the structure of interest, ε nuisance independent of the class, and n_y nuisance that varies with the class. Averaging many examples of a class drives ε towards zero, whereas n_y is present in every example by construction, so the two regimes call for different operations and the measurements below are chosen to tell them apart.

Representations. MERT-v1-330M [2] layers 4 to 24, EnCodec 32 kHz [11], the codec MusicGen consumes [12], and a log-CQT [8, 9] as an input floor. A chromagram serves as the music-theoretic positive control. Clips are pooled to a single vector by a mean over frames.

Key-centroid geometry. On GiantSteps [13], with 7035 clips covering all 24 keys, we average clip embeddings within key to obtain 24 centroids and rank-correlate their pairwise Euclidean distances against three reference structures: circle-of-fifths distance, the Krumhansl–Kessler key-profile distance, and semitone distance. Significance uses a 2000-draw permutation null over key labels, and a partial Spearman additionally removes mel-spectral centroid distance. The semitone reference is reported alongside, because a generic tendency for nearby keys to look alike would raise both.

Readout geometry. The same probe, a multinomial logistic regression under 5-fold cross-validation, yields 24 class weight vectors, and we apply the identical measurement to their pairwise cosine distances.

Octave equivalence. On NSynth [14] we use real recordings only, since pitch shifting by signal processing would build the answer into the stimulus. For each instrument we fix an anchor set of pitches supporting every transposition up to 25 semitones, so the anchor set is constant across transpositions, and measure the mean cosine similarity between a note and its transposition by k semitones. Three statistics are reported together, as registered in advance: a local octave contrast, comparing the similarity at 12 semitones against the mean of 11 and 13; a strict contrast, comparing it against the largest similarity anywhere from 8 to 11 semitones; and the coefficient on a twelve-semitone periodic term in a fitted curve. Confidence intervals are bootstrapped over *instruments*. We require the three to agree in sign, since a positive periodic coefficient with no local peak at 12 semitones is a curve-fitting artefact.

Subspaces. Each subspace is an orthonormal basis of the top d shrinkage-LDA discriminant directions for a stated target, with d at most 11, since any linear map to a twelve-class target has rank at most 11. We report the fraction of total variance the subspace holds and, as a control, matched random orthonormal subspaces, because projection changes the metric whatever the directions are.

4. RESULTS

4.1. Information, metric and readout dissociate

Table 1 measures all three on the same clips and keys, and the control block sets the scale for reading it. An analytic circle of fifths scores close to 1 in every column, so a null elsewhere is a null of the representation and not of the method, while random features show nothing anywhere. The load-bearing control is the third one, in which orthogonal per-key codes are equidistant by construction. They decode key at .651, exceeding the accuracy of every real representation, and yet they show no fifths geometry in either their centroids, at -0.008, or their readout weights, at -0.005. This follows the logic of control tasks [4] without relying on matched accuracy, and it rules out the natural sceptical reading, that keys a fifth apart are confusable whatever was fitted, so any accurate 24-way classifier ends up with fifths-structured weights.

Table 1. Key information, ambient geometry and readout geometry come apart. Same 7035 GiantSteps clips, same 24 keys. *acc* is 24-way key accuracy (chance .042); ρ_5^{cent} is the circle-of-fifths correlation among the 24 key centroids, i.e. the ambient metric; ρ_5^{W} is the same correlation among the probe's 24 class weight vectors, i.e. what a linear readout reaches. The controls fix the scale for reading the two geometry columns.

representation	acc	ρ_5^{cent}	ρ_5^{W}
Controls			
analytic fifths (metric ctrl)	0.274	$+0.977$	$+0.986$
random features (no info)	0.055	$+0.009$	$+0.033$
orthogonal key codes (no tonal rel.)	0.651	-0.008	-0.005
Spectral front ends			
log-CQT, 84 bin, dB	0.492	-0.035	$+0.560$
log-CQT, 84 bin, per-frame norm	0.486	-0.117	$+0.671$
octave-summed, 12 bin, dB	0.407	-0.106	$+0.305$
chromagram (fold + norm)	0.451	$+0.475$	$+0.692$
Neural codec			
EnCodec 32 kHz	0.404	$+0.177$	$+0.648$
MERT-v1-330M			
layer 4	0.451	$+0.445$	$+0.412$
layer 12	0.456	$+0.468$	$+0.428$
layer 16	0.440	$+0.395$	$+0.409$
layer 24	0.418	$+0.294$	$+0.484$

The dissociation that follows is unambiguous, in that the log-CQT has the highest key accuracy of any real arm and no fifths structure in its centroids, at -0.035 with p of .52, while its probe weights carry that structure strongly, at 0.560 with a z of 9.1. Tonal relations are therefore linearly accessible in a log-spectrogram and simply absent from its metric, which requires no learned encoder at all.

What MERT does is move structure between columns rather than create it. Its readout geometry, between 0.41 and 0.48, is *lower* than that of the log-CQT, EnCodec or the chromagram, while its centroid geometry is far higher. Across depth, tonal relational geometry weakens, from 0.445 at layer 4 to 0.294 at layer 24, while major and minor become more separable, from 0.216 to 0.328. We report this as a correlation rather than a mechanism, since key accuracy does not improve along the way, and the mode effect is -0.054 and not significant in the chromagram control, so the growing mode axis is not pitch-class based and is at least as consistent with

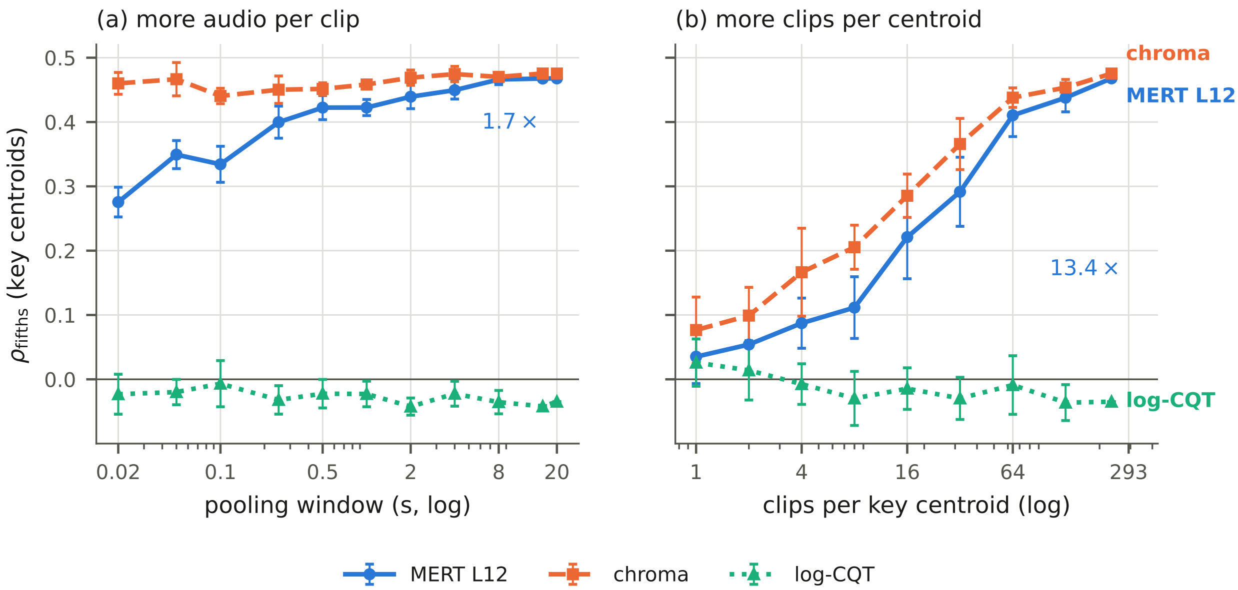

Fig. 1. Clip-level key geometry is set by centroid sample size, not by temporal context. Both panels plot the same quantity on the same vertical axis. (a) A thousandfold longer pooling window moves MERT L12 by a factor of 1.7 and moves the chromagram by 3%, and a single 13 ms frame already carries 59% of the 20 s value. (b) Increasing the number of clips averaged into each key centroid moves it by a factor of 13.4, and in the single-example centroid regime the geometry is absent. Error bars are one standard deviation over 5 draws in (a) and 10 in (b).

production and instrumentation in an all-EDM corpus.

4.2. Sample-limited masking

The clip-level result invites a temporal reading, in which keys are properties of seconds of music, but that reading does not survive the measurement. Fig. 1(a) sweeps the pooling window from one frame to the full 20 s clip, drawing one window per clip at every length so the number of clips behind each centroid is held constant. A single 13 ms MERT frame already yields a correlation of 0.276, which is 59% of the 20 s value, and the chromagram is flat to within 3%, with no knee and no threshold anywhere in the sweep.

The clip-level measurement averages twice, and only one of those averages was being varied. Fig. 1(b) holds the window at the full clip and sweeps the number of clips per centroid, giving 0.035 at one clip and 0.468 at all of them for MERT L12, a factor of 13.4 against 1.7 for a thousandfold longer window. In the single-example centroid regime, fifths geometry is indistinguishable from zero for every arm.

This is Eq. (1) with ε dominant, because what differs between two GiantSteps tracks in the same key, namely production, instrumentation and subgenre, is largely unrelated to the key and cancels in the mean. Centroid sample size is therefore the dominant source of the apparent clip-level effect, and while a longer window is not entirely without influence, it accounts for a small fraction of the same difference.

4.3. Metric masking

If low between-item signal-to-noise were the whole story, the same averaging should expose octave equivalence on NSynth, and we pre-registered that prediction in advance of the run that refutes it. Across a sweep over notes per pitch centroid, every arm saturates by four notes at approximately zero, MERT L24 remains negative, and the chromagram needs no averaging at all, reaching 0.566 from single notes. Register and spectral envelope are functions of pitch, so they survive into every pitch centroid, and averaging over instruments cancels the instrument but never the octave, which is the behaviour Eq. (1) attributes to n_y.

A projection can nonetheless expose the target relation despite n_y, and the evidence here is cross-dataset and cross-task. We fit LDA directions on GiantSteps track tonics only, with no notes, no NSynth and no octave information, and apply them unchanged to individual NSynth recordings with no averaging. At rank 4 the strict octave contrast moves from -0.005 to 0.223 for MERT L12, from -0.001 to 0.304 for layer 4, and from 0.004 to 0.466 for the log-CQT, with the other two statistics agreeing in sign and bootstrap intervals excluding zero. Fig. 2(c) shows the curve those contrasts are drawn from, where the projected similarities reach maxima at exactly twelve and twenty-four semitones against an ambient baseline that decays smoothly. Matched random subspaces stay between -0.019 and 0.007. The subspace uses between 0.11 and 0.24% of MERT's variance and 1.85% of the log-

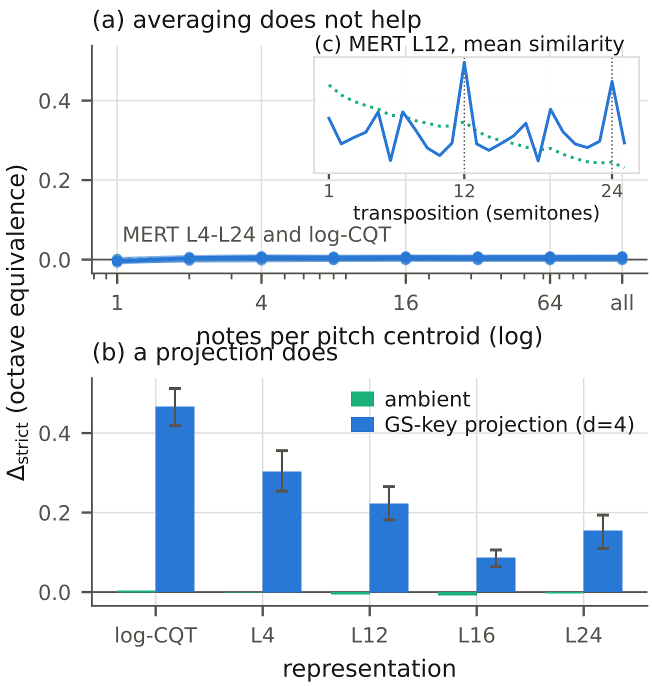

Fig. 2. Two operations on NSynth notes, with one outcome each. Panels (a) and (b) share a vertical axis, since rescaling (a) would invent structure that is not there. (a) Averaging notes of the same pitch into a pitch centroid does not produce octave equivalence at any sample size, with every MERT layer and the log-CQT flat at zero and MERT L24 remaining negative. The chromagram is off scale throughout, moving only from 0.57 to 0.62, because it has octave equivalence already in single notes. (b) A rank-4 projection fitted only on GiantSteps track tonics, applied unchanged to individual recordings, with instrument-bootstrap 95% intervals. (c) The curve (b) summarises, for MERT layer 12 and in the colours of (b), standardised within each condition.

CQT's.

It would overstate the case to say key supervision *forces* pitch-height invariance, since GiantSteps key correlates with subgenre, tempo and production, and an LDA may use any of it. What the transfer licenses is narrower: structure that survives the move to an independently recorded isolated-note corpus cannot be solely a GiantSteps-specific nuisance correlation. Nor is this a mechanistic claim about the projection, since what the transfer establishes is that a low-rank reweighting exposes octave-equivalent structure, and whether it does so by suppressing class-coupled variation or by amplifying the tonal component is not identifiable from this measurement.

A registered prediction that failed. We predicted that removing the subspace from the representation would at least halve clip-level fifths geometry. Against a baseline recomputed on the same held-out clips it does not move it, going from 0.127 to 0.119 for MERT L12 and from 0.118 to 0.114 for layer 4, and only the twelve-dimensional chromagram collapses, which is a dimensionality effect. A subspace holding well under 1% of the variance is therefore sufficient to expose the relation, but the relation is not localised exclusively to the one we recover. How widely it is distributed would need repeated recovery after sequential deflation, which we do not attempt here.

5. DISCUSSION

Information availability, metric visibility and readout accessibility are separable, and separating them changes what a probe result licenses. A high probe accuracy says the structure is linearly recoverable and says nothing about whether it governs the distances a model relying on similarity or retrieval consumes.

The two masking modes suggest different remedies. Sample-limited masking is a property of the evaluation, and the fix is more examples per class, not longer audio. Reporting the number of items behind each centroid should be standard, since the same encoder scores 0.035 or 0.468 depending only on that. Metric masking is a property of the representation itself, which no evaluation protocol removes and which a learned transform can work around.

Nothing in Eq. (1) is specific to music. Semantic relations in vision may equally be linearly available while the Euclidean geometry is dominated by pose or background, and phonetic structure in speech may be masked by speaker and channel, so the distinction plausibly extends beyond music representations.

Limitations. GiantSteps is entirely electronic dance music, where key correlates with subgenre. The cross-dataset transfer in Sec. 4.3 constrains that confound but does not remove it, and a key-balanced, genre-diverse corpus would settle it. NSynth is monophonic and drawn from isolated instruments, so its nuisance structure is unusually clean. Our subspaces are linear by design, and a nonlinear transform might expose more at the cost of interpretability. Finally, the mode effect is reported but not explained, and we do not claim it is harmonic.

6. CONCLUSION

Musical structure can be present in a representation, linearly accessible and still invisible in its geometry, for two reasons calling for two different operations. Averaging over examples accounts for most of the apparent clip-level key geometry, while a rank-4 projection fitted on another dataset and task exposes octave equivalence that no amount of averaging reaches. Neither operation creates the structure it exposes, and knowing which of the two a null calls for is what separating them buys.

7. REFERENCES

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